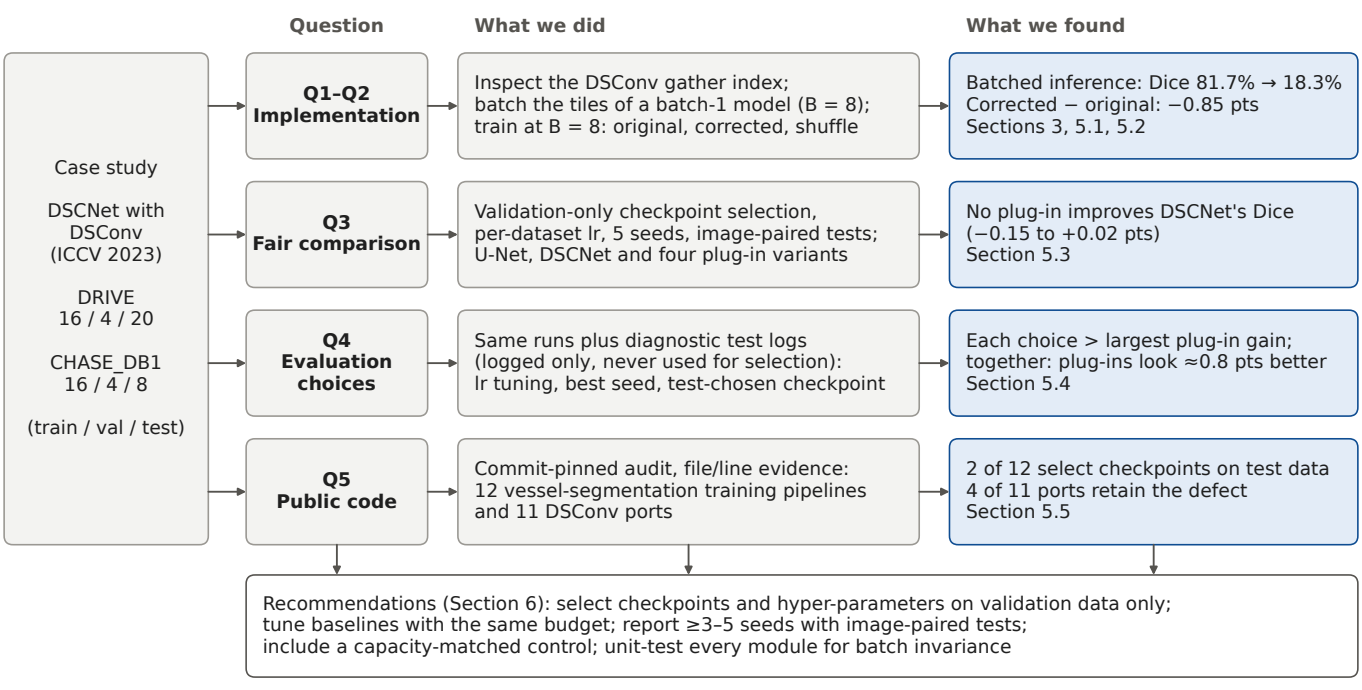
Question
What we did
What we found
Case study
DSCNet with DSConv (ICCV 2023)
DRIVE
16 / 4 / 20
CHASE_DB1
16 / 4 / 8
(train / val / test)
Q1–Q2
Implementation
Inspect the DSConv gather index;
batch the tiles of a batch-1 model (B = 8);
train at B = 8: original, corrected, shuffle
Batched inference: Dice 81.7% → 18.3%
Corrected − original: −0.85 pts
Sections 3, 5.1, 5.2
Q3
Fair comparison
Validation-only checkpoint selection,
per-dataset lr, 5 seeds, image-paired tests;
U-Net, DSCNet and four plug-in variants
No plug-in improves DSCNet's Dice
(−0.15 to +0.02 pts)
Section 5.3
Q4
Evaluation choices
Same runs plus diagnostic test logs
(logged only, never used for selection):
lr tuning, best seed, test-chosen checkpoint
Each choice > largest plug-in gain;
together: plug-ins look ≈0.8 pts better
Section 5.4
Q5
Public code
Commit-pinned audit, file/line evidence:
12 vessel-segmentation training pipelines
and 11 DSConv ports
2 of 12 select checkpoints on test data
4 of 11 ports retain the defect
Section 5.5
Recommendations (Section 6): select checkpoints and hyper-parameters on validation data only;
tune baselines with the same budget; report ≥3–5 seeds with image-paired tests;
include a capacity-matched control; unit-test every module for batch invariance